

Crack Mapping Report — Pier P2 Pier Cap

Nandira River Bridge Project — Package NRB-4

Document Reference:	CMR-NRB4-001
Date:	15-Mar-2024
Contract No.:	NHIA/NRB4/CW/2020-01
From:	Suvarna Materials Testing Laboratory
To:	Meridian Engineering Consultants
CC:	Sagara Constructions Pvt. Ltd.

Subject: Crack Mapping Survey — Pier P2 Pier Cap Top Surface

Crack mapping survey carried out 11-Mar-2024, against the 1m x 1m reference grid established in Joint Site Inspection Record SIR-NRB4-001.

SURVEY RESULTS:

Zone	Crack Density	Width Range	Pattern
Central (grid C3-D5)	High — interconnected	0.2mm – 0.5mm	Map/crazing
Intermediate (grid B2-B6, E2-E6)	Moderate	0.1mm – 0.3mm	Map/crazing
Edge (grid A1-A7, F1-F7)	Low, isolated cracks only	0.1mm – 0.15mm	Isolated fine

1. PATTERN: The cracking forms an irregular, interconnected map (or 'crazing') pattern, with no dominant single crack, no radial pattern consistent with a point load, and no orientation aligned with the reinforcement layout drawings reviewed for this survey. Crack density decreases from the centre of the pour toward its edges.
2. EXTENT: Cracking is confined to the top surface of the pier cap. No crack was traced to the pier cap soffit or the vertical faces of the pier shaft during this survey.
3. ASSESSMENT: This pattern, density gradient, and confinement to the top (exposed) surface are characteristic of surface shrinkage cracking (plastic or early-age drying shrinkage), rather than structural or load-induced cracking, which would typically show a different orientation and would not be confined to a single exposed surface. Depth of cracking to be confirmed by core testing and petrographic examination, reported separately.

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